

Topic 03: Existence and Uniqueness of Solutions

Professor Leonard Master Series · Lesson 11

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: Does a Solution Even Exist? And is it the Only One?

Professor Leonard always stresses the two foundational questions of mathematics:

1. **Existence:** Does at least one solution curve pass through the point (x_0, y_0) ?
2. **Uniqueness:** Is that solution curve

the only one

, or do curves fork and branch out into multiple realities?

In physics and engineering, uniqueness is non-negotiable: if you release a pendulum with exact initial position and velocity, the universe doesn't pick between two different futures!

2. The Picard-Lindelöf Existence & Uniqueness Theorems

Theorem A: First-Order Linear Equations (Global on Interval)

Consider the linear IVP in standard form:

$$y' + P(x)y = Q(x), \quad y(x_0) = y_0$$

Rule: If $P(x)$ and $Q(x)$ are continuous on an open interval $I = (a, b)$ containing x_0 , then there exists **one and only one** solution $y(x)$ valid across the **entire interval** I .

- The interval of existence is determined **solely by the coefficients** $P(x)$ and $Q(x)$, before you even solve the ODE!

Theorem B: First-Order Non-Linear Equations (Local Theorem)

Consider the IVP:

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0$$

Let R be an open rectangle $a < x < b, c < y < d$ containing (x_0, y_0) .

1. **Existence:** If $f(x, y)$ is continuous on R , then there exists
at least one

solution in an interval $(x_0 - h, x_0 + h)$.

2. **Uniqueness:** If **both** $f(x, y)$ **and** its partial derivative $\frac{\partial f}{\partial y}$ are continuous on R , then the solution is **unique** on that interval.

3. Professor Leonard's Whiteboard Problem Walkthroughs

Problem 3.1: The Classic Failure of Uniqueness (The Cusp Fork)

Statement: Consider the IVP:

$$\frac{dy}{dx} = 3y^{2/3}, \quad y(0) = 0$$

1. Check the conditions of the Existence & Uniqueness Theorem at $(0, 0)$.
2. Demonstrate that this IVP possesses infinitely many distinct solutions!

Step-by-Step Whiteboard Solution:

- **Step 1: Check continuity of $f(x, y)$:**

$$f(x, y) = 3y^{2/3}$$

$f(x, y)$ is continuous everywhere in the xy -plane. Therefore, **existence of a solution is guaranteed!**

- **Step 2: Check continuity of $\frac{\partial f}{\partial y}$:**

$$\frac{\partial f}{\partial y} = 3 \cdot \frac{2}{3} y^{-1/3} = \frac{2}{y^{1/3}} = \frac{2}{\sqrt[3]{y}}$$

At $y = 0$, the denominator is 0! $\frac{\partial f}{\partial y}$ **is discontinuous (blows up to ∞) along the entire x -axis ($y = 0$)!**

The Uniqueness condition fails at $(0, 0)$. Uniqueness is **NOT** guaranteed.

- **Step 3: Find explicit multiple solutions:**

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Candidate 1 (Trivial Solution)

: $y_1(x) \equiv 0$. $y_1'(x) = 0$, and $3(0)^{2/3} = 0$. $y_1(0) = 0$. Valid!

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Candidate 2 (Separation of Variables)

:

$$y^{-2/3} dy = 3 dx \implies 3y^{1/3} = 3x + C \implies y^{1/3} = x + \frac{C}{3}$$

Apply $y(0) = 0 \implies 0 = 0 + C/3 \implies C = 0$.

$$y^{1/3} = x \implies y_2(x) = x^3$$

$y_2'(x) = 3x^2$. Also $3(x^3)^{2/3} = 3x^2$. $y_2(0) = 0$. Valid!

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Candidate 3 (Branching Infinite Family)

:

$$y_c(x) = \begin{cases} 0, & x \leq c \\ (x - c)^3, & x > c \end{cases} \quad (\text{for any } c \geq 0)$$

All satisfy $y(0) = 0$! A particle resting at $(0, 0)$ can choose to sit still forever, or branch off along a cubic curve at any arbitrary time c !

Problem 3.2: Finding the Maximum Guaranteed Interval for a Linear IVP

Statement: Without solving the equation, determine the largest interval in which the solution to the following IVP is guaranteed to exist and be unique:

$$(x^2 - 9) \frac{dy}{dx} + 2xy = \ln(x + 1), \quad y(2) = 5$$

Step-by-Step Whiteboard Solution:

- **Step 1: Put the equation into standard linear form $y' + P(x)y = Q(x)$:** Divide through by the leading coefficient $(x^2 - 9)$:

$$\frac{dy}{dx} + \frac{2x}{x^2 - 9}y = \frac{\ln(x + 1)}{x^2 - 9}$$

- **Step 2: Identify $P(x)$ and $Q(x)$ and their domains:**

$$P(x) = \frac{2x}{(x - 3)(x + 3)}$$

Continuous everywhere except $x = -3$ and $x = 3$.

$$Q(x) = \frac{\ln(x + 1)}{(x - 3)(x + 3)}$$

Requires $x + 1 > 0 \implies x > -1$, and denominator $\neq 0 \implies x \neq 3$.

- **Step 3: Find the intersection of continuity for both $P(x)$ and $Q(x)$:**
 - $x > -1$ and $x \neq 3$.
 - Continuous intervals are: $I_1 = (-1, 3)$ and $I_2 = (3, \infty)$.
- **Step 4: Check where the initial condition $x_0 = 2$ lives:** $x_0 = 2 \in (-1, 3)$.
- **Conclusion:** The theorem guarantees a unique solution on the interval:

$$I = (-1, 3)$$

Problem 3.3: Non-Linear Finite-Time Blow-Up (Why Nonlinear Intervals Depend on y_0)

Statement: Solve $y' = y^2$, $y(0) = y_0$. Show that unlike linear ODEs, the interval of existence depends directly on the initial value y_0 .

Step-by-Step Whiteboard Solution:

- **Step 1: Separate and integrate:**

$$y^{-2} dy = dx \implies -\frac{1}{y} = x + C \implies y(x) = \frac{1}{-x - C} = \frac{1}{C_1 - x}$$

- **Step 2: Apply** $y(0) = y_0$:

$$y_0 = \frac{1}{C_1} \implies C_1 = \frac{1}{y_0}$$

$$y(x) = \frac{1}{\frac{1}{y_0} - x} = \frac{y_0}{1 - y_0 x}$$

- **Step 3: Analyze the domain of definition:** The solution blows up to ∞ at vertical asymptote $x = 1/y_0$.
 - If $y_0 = 1$, the solution exists only on $(-\infty, 1)$.
 - If $y_0 = 100$, it blows up at $x = 1/100 = 0.01$! Notice that $f(x, y) = y^2$ is smooth everywhere, but the solution cannot survive past $x = 1/y_0$!

Problem 3.4: Discontinuous Leading Coefficient & The Zero-Crossing Trap

Statement: Find all solutions to $x \frac{dy}{dx} = y, y(0) = 0$.

Step-by-Step Whiteboard Solution:

- **Step 1: Separate for** $x \neq 0, y \neq 0$:

$$\frac{dy}{y} = \frac{dx}{x} \implies \ln |y| = \ln |x| + C \implies y(x) = kx \quad (k \in \mathbb{R})$$

- **Step 2: Apply** $y(0) = 0$: $0 = k(0)$ is identically true for EVERY real number k ! Every line $y = kx$ passing through the origin is a solution!
- **Step 3: Why did uniqueness fail?**: In standard form, $y' - \frac{1}{x}y = 0$, $P(x) = -1/x$ is undefined and discontinuous at $x = 0$.

4. Common Exam Traps & Professor Leonard Warnings

- **Trap 1: Forgetting Standard Form:** If $(x - 1)y' + y = 0$, you must divide by $(x - 1)$ before checking continuity!
- **Trap 2: Ignoring Domain of ln and Roots:** $\ln(x)$ requires $x > 0$; $\sqrt{x}$ requires $x \geq 0$.
- **Trap 3: Assuming Non-Linear Interval is Constant:** In nonlinear ODEs, you cannot find the interval of existence without solving the IVP first.